

Advanced Cognitive Neuroscience

Week 37: Workshop paradigm: Measuring visual subjective experience + MR Recordings

You

- My aim:
 - Every time I address you, I will do it with your name
 - Some times I will be wrong, and that might cause some humiliation both ways
 - Please correct me when I am wrong

Me as a lecturer

Me asking questions

- I will every now and then pick a student at random to answer questions
- I want to emphasise that I am only picking students that I judge are paying attention
 - i.e.: I am not trying to humiliate students not paying attention
- It is completely okay to say “I do not have an answer”; then I’ll just pass it on to someone else

You asking questions

- Does such a thing as a stupid question exist?
- I don't care if they do or not
- You ask a question because you want to know ...
- ... and I answer because I want to tell you

Linear algebra materials

<https://www.3blue1brown.com/>

Look for the linear algebra videos

Leon Steven J (2015) Linear algebra with applications.,
9. ed., global ed. Pearson, Boston ;

Workshop

SUBJECTIVE EXPERIENCE

- September 16th: MR scans (**only** the subjects should show up):
- September 8-10: 9-17
 - Two subjects the first day
 - Three subjects the other days
 - Subjects can join one other group to observe how things are done from the outside

MEG slots

T u e s d a y 8	G r o u p n u m b e r
1 0 . 0 0 - 1 2 . 4 5	4
1 2 . 4 5 - 1 5 . 3 0	1

W e d n e s d a y 9	G r o u p n u m b e r
9 . 0 0 - 1 1 . 4 5	7
1 1 . 4 5 - 1 4 . 3 0	8
1 4 . 3 0 - 1 7 . 1 5	5

T h u r s d a y 1 0	G r o u p n u m b e r
9 . 0 0 - 1 1 . 4 5	3
1 1 . 4 5 - 1 4 . 3 0	2
1 4 . 3 0 - 1 7 . 1 5	6

MR slots

September 16th

you will receive more info

Study groups

Study group no	Full name
1	Denitsa Delyanova Pehlivanova
1	Akaran Sivakumar
1	Anne Christine Vig Jensen
1	Christian Vejen Barnkob
1	Valeria Alladio
2	Vyara Krasteva
2	Alexander Juby-Rasmussen
2	Daniel Lindblad Sørensen
2	Nadia Kinga Wójtowicz
3	Anders Herzog Varan
3	Andreas Møldrup Holst
3	Clara Holst
3	Zuzanna Katarzyna Zyla
3	Cosma Carlotta Rost
4	Gustav Idun Sloth
4	Halldan Nordahl Fundal
4	Ida Elmo Brøcker
4	Paula Samide
4	Lilla Fürtön

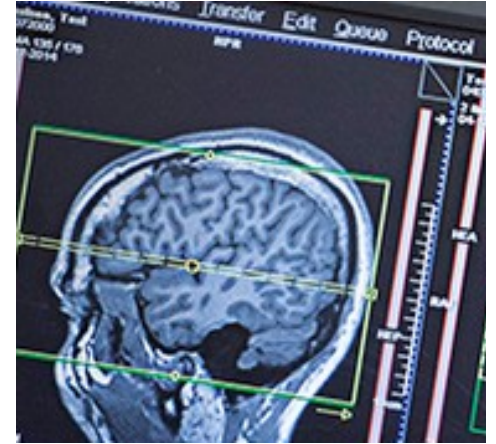
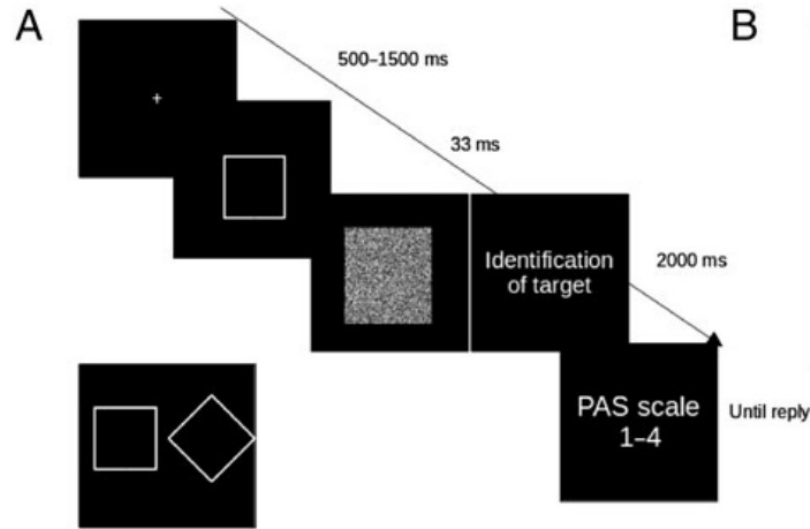
5	Johannes Eide Rambøll
5	Laura Emilie Lundbye
5	Louie Mølgaard Hessellund
5	Rose Lina Reinhold
5	Reka Forgo
6	Frederikke Lykke Korreborg
6	Liv Drasbek
6	Lucas Johannes Simonsen
6	Lydia Pauli Rambrand
6	Anne Skamris Holm
7	Marie Vestergaard Thomsen
7	Mikkel Lindhardt Glavind Andersen
7	Nanna Klitgaard Damgaard
7	Niels Engedal
8	Sara Eline Røeder Nielsen
8	Sissel Højgaard Vang-Pedersen
8	Vilma Bejer Kristiansen
8	Julianne Silke Swärd

Workshop

SUBJECTIVE EXPERIENCE



from Center of Functionally
Integrative Neuroscience



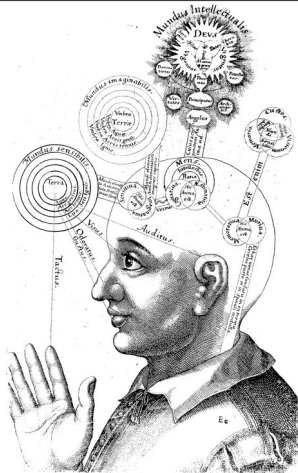
Andersen LM, Pedersen MN, Sandberg K, Overgaard M (2016)
Occipital MEG Activity in the Early Time Range (<300 ms)
Predicts Graded Changes in Perceptual Consciousness.
Cerebral Cortex 26:2677–2688.
<https://doi.org/10.1093/cercor/bhv108>

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Workshop

Table 1 The Perceptual Awareness Scale (PAS)

Label	Description [from Ramsøy and Overgaard (2004)]
(1) No Experience (NE)	No impression of the stimulus. All answers are seen as mere guesses.
(2) Weak Glimpse (WG)	A feeling that something has been shown. Not characterized by any content, and this cannot be specified any further.
(3) Almost Clear Experience (ACE)	Ambiguous experience of the stimulus. Some stimulus aspects are experienced more vividly than others. A feeling of almost being certain about one's answer.
(4) Clear Experience (CE)	Non-ambiguous experience of the stimulus. No doubt in one's answer.



Andersen LM, Pedersen MN, Sandberg K, Overgaard M (2016) Occipital MEG Activity in the Early Time Range (<300 ms) Predicts Graded Changes in Perceptual Consciousness. *Cerebral Cortex* 26:2677–2688.
<https://doi.org/10.1093/cercor/bhv108>

Ramsøy TZ, Overgaard M (2004) Introspection and subliminal perception. *Phenomenology and the Cognitive Sciences* 3:1–23.
<https://doi.org/10.1023/B:PHEN.0000041900.30172.e8>

```
##% RUN EXPERIMENT
```

```
## initialise
experiment = Experiment()

## setting up experiment
experiment.open_GUI()
experiment.set_experiment_parameters()
experiment.check_user()
experiment.write_to_terminal('refresh_rate')
experiment.define_io_files()
experiment.write_to_terminal('setting_path')
experiment.define_texts()
experiment.create_experiment_window()
experiment.define_visual_stimuli()
experiment.present_instructions('welcome')

## practice
experiment.present_instructions('practice')
experiment.run_practice()

## experiment
experiment.set_experiment_parameters()
experiment.present_instructions('experiment')
experiment.run_experiment()

## thank you
experiment.present_instructions('thank_you')
```

Check out the experiment on your own computer

https://github.com/ualsbombe/2026_advanced_cognitive_neuroscience/tree/main/week37/MEG_experiment



Protocol

https://github.com/ualsbombe/2026_advanced_cognitive_neuroscience/blob/main/week37/MEG_experiment/data_acquisition_protocol.html

Context of the experiment

State consciousness



Content consciousness



Aim:

Answering where and when in the brain, we can see neural activity correlating with *consciousness of visual content*

NCC (Neural Correlate of Consciousness)

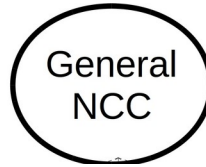
“An NCC (for content) is a minimal neural representational system N such that representation of a content in N is sufficient, under conditions C, for representation of that content in consciousness.”

Chalmers, D. J. (2000). What Is a Neural Correlate of Consciousness. In T. Metzinger (Ed.), Neural Correlates of Consciousness

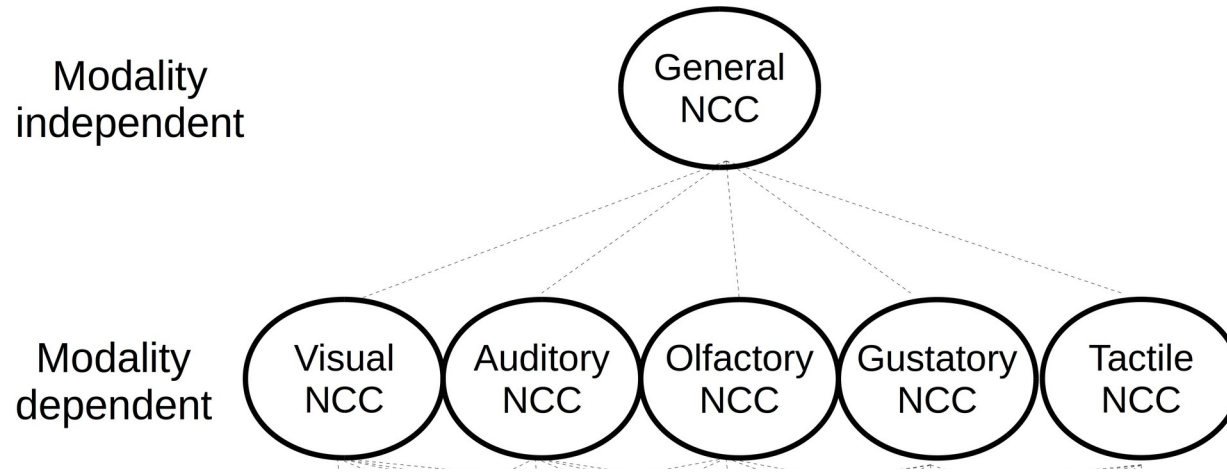
CC BY Licence 4.0: Lau Møller Andersen 2026

Levels of consciousness

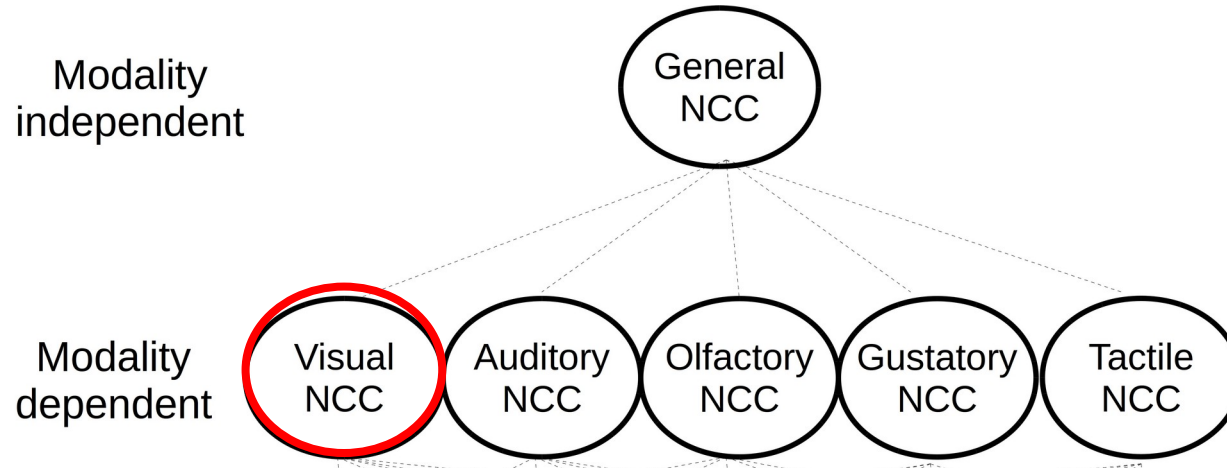
Modality
independent



Levels of consciousness

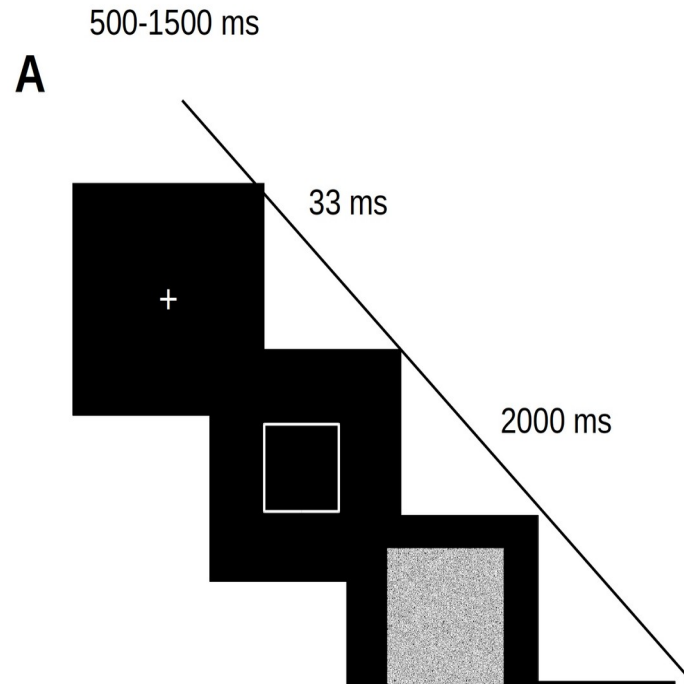


Levels of consciousness

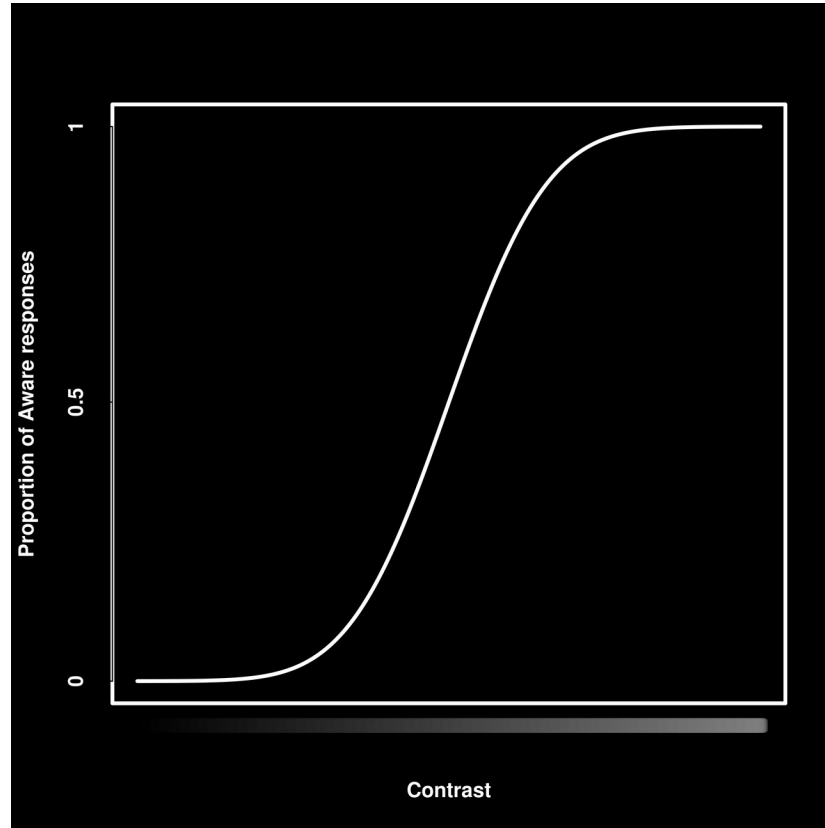


Paradigm and basic MEG

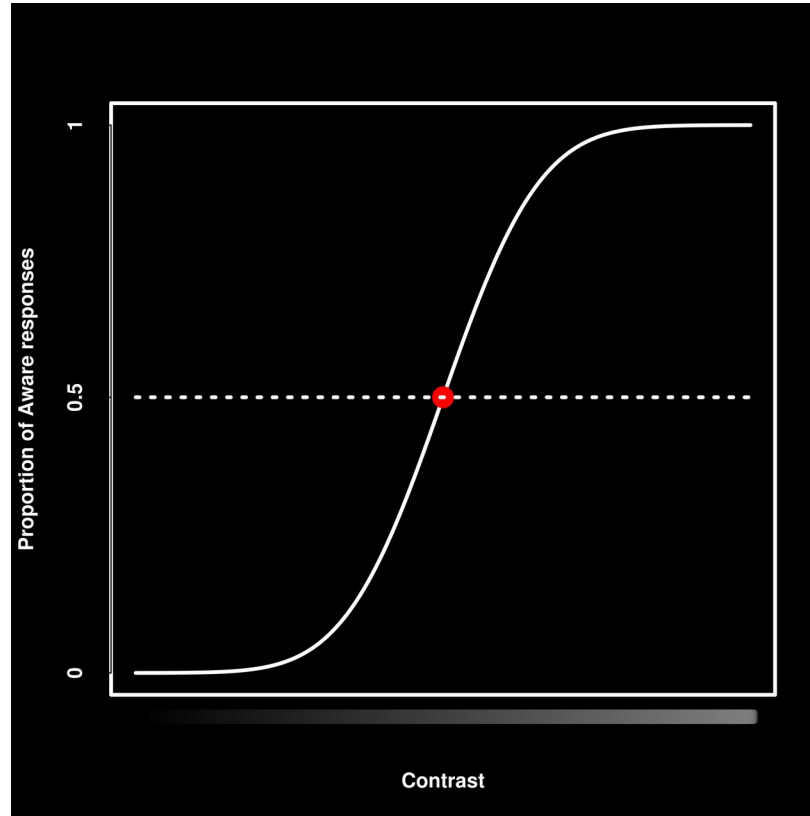
Visual masking



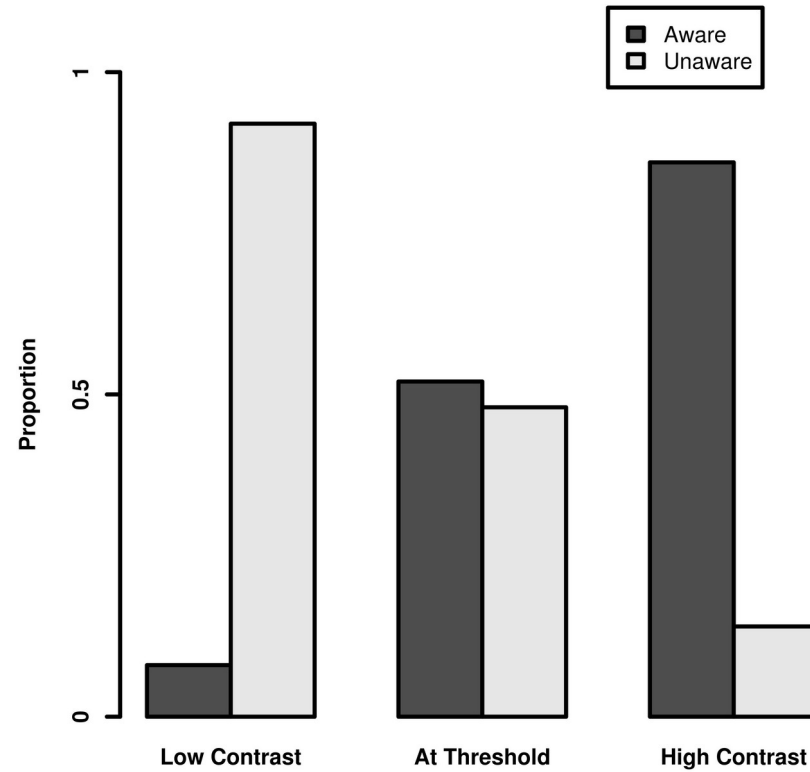
Threshold



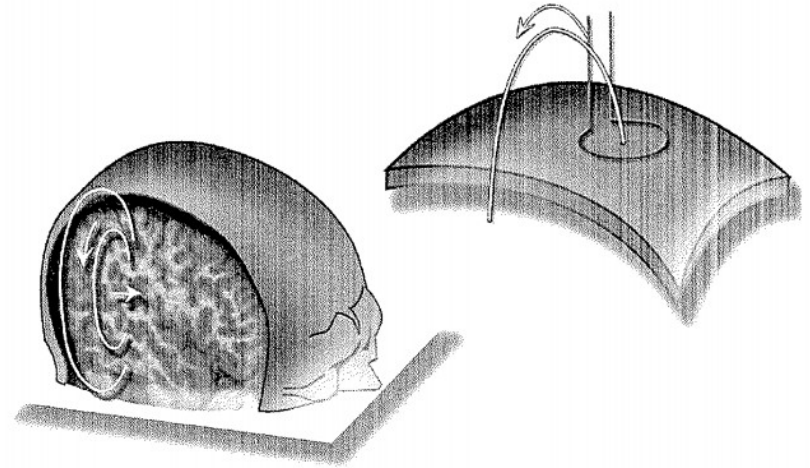
Threshold



Threshold

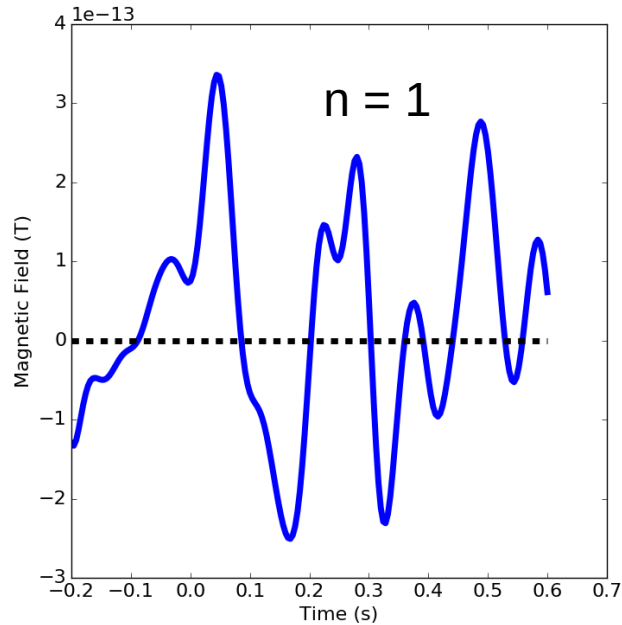


Magnetoencephalography

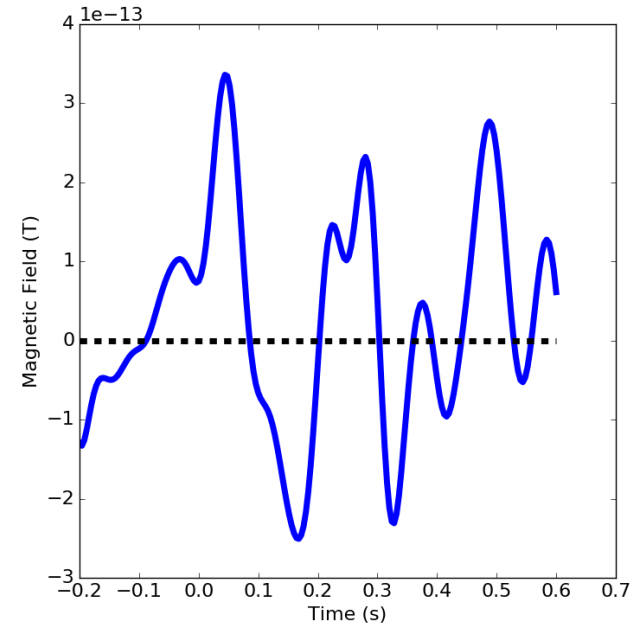


*Papanicolaou 2009
Clinical Magnetoencephalography and Magnetic
Source Imaging.*

Epochs and Evoked Responses

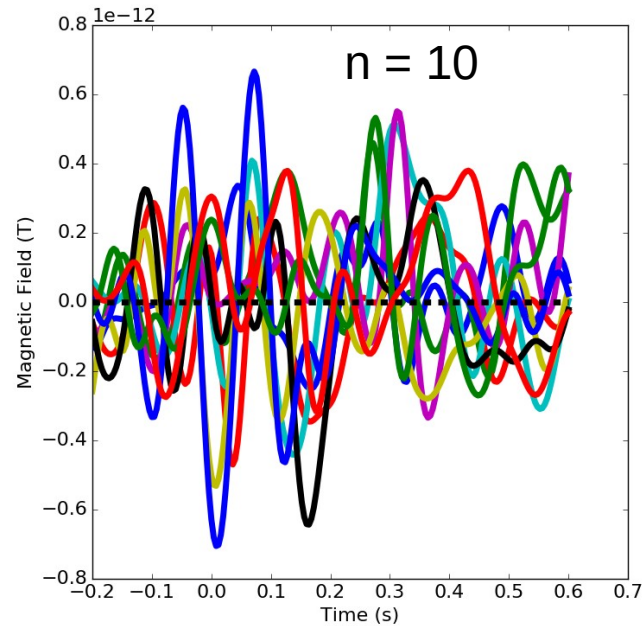


EPOCHS

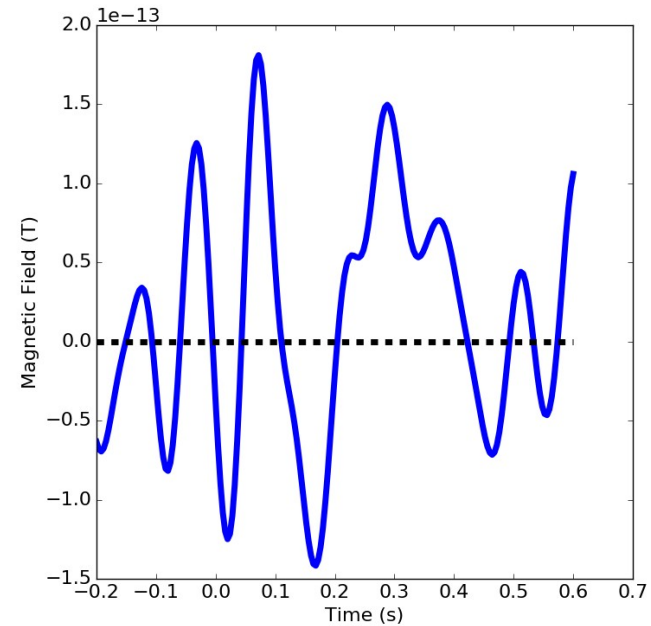


EVENT RELATED FIELD
(ERF)

Epochs and Evoked Responses



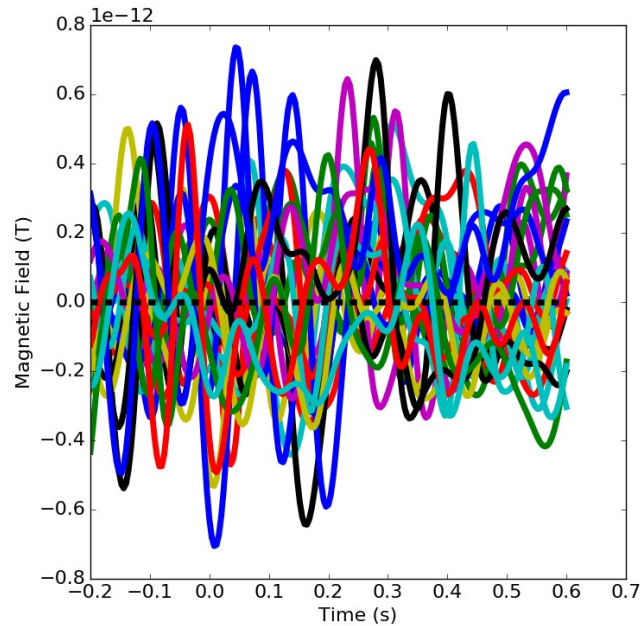
EPOCHS



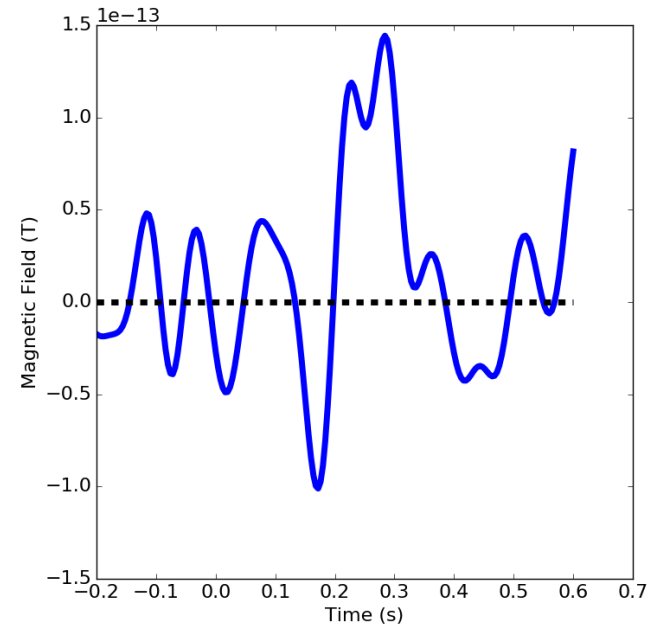
EVENT RELATED FIELD
(ERF)

Epochs and Evoked Responses

$n = 25$



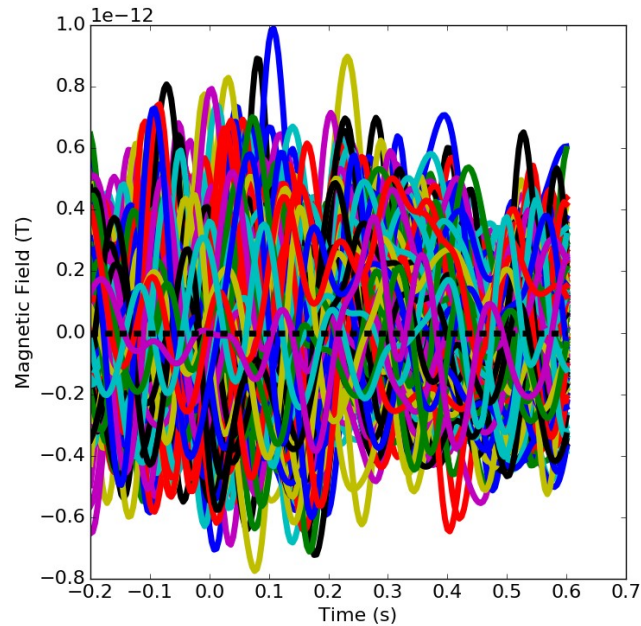
EPOCHS



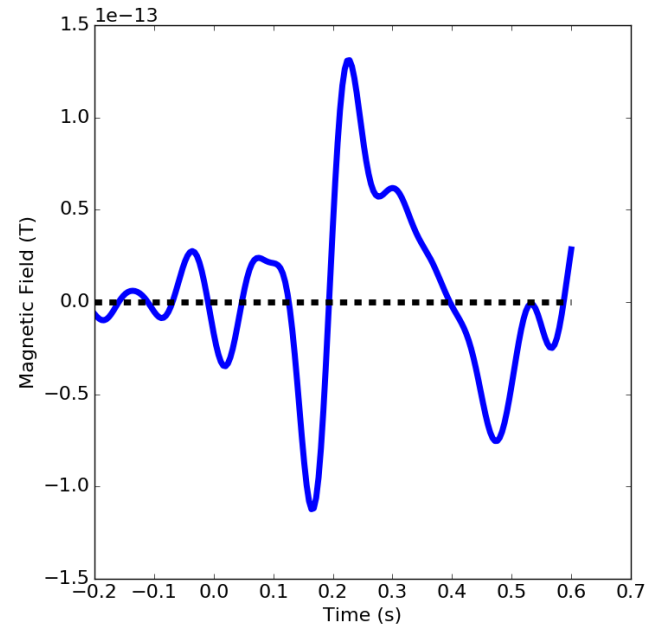
EVENT RELATED FIELD
(ERF)

Epochs and Evoked Responses

$n = 250$



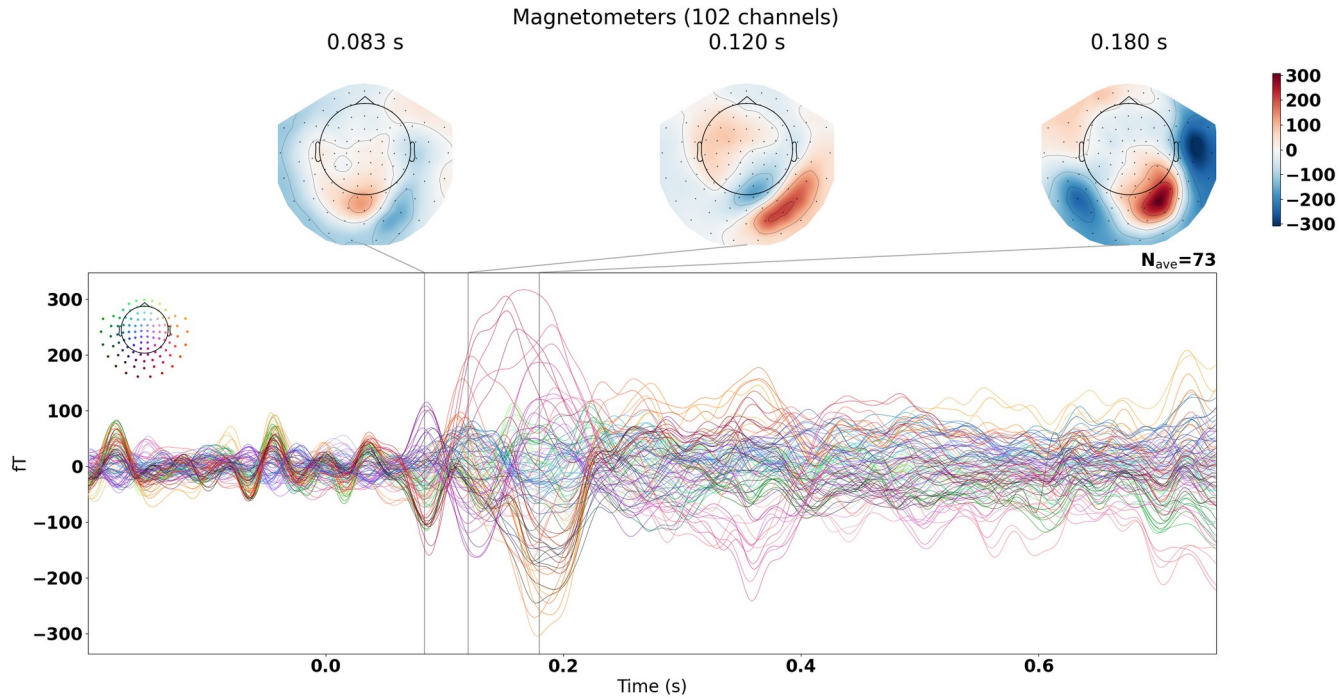
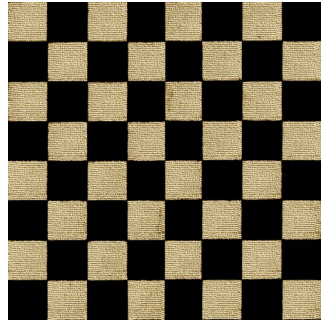
EPOCHS



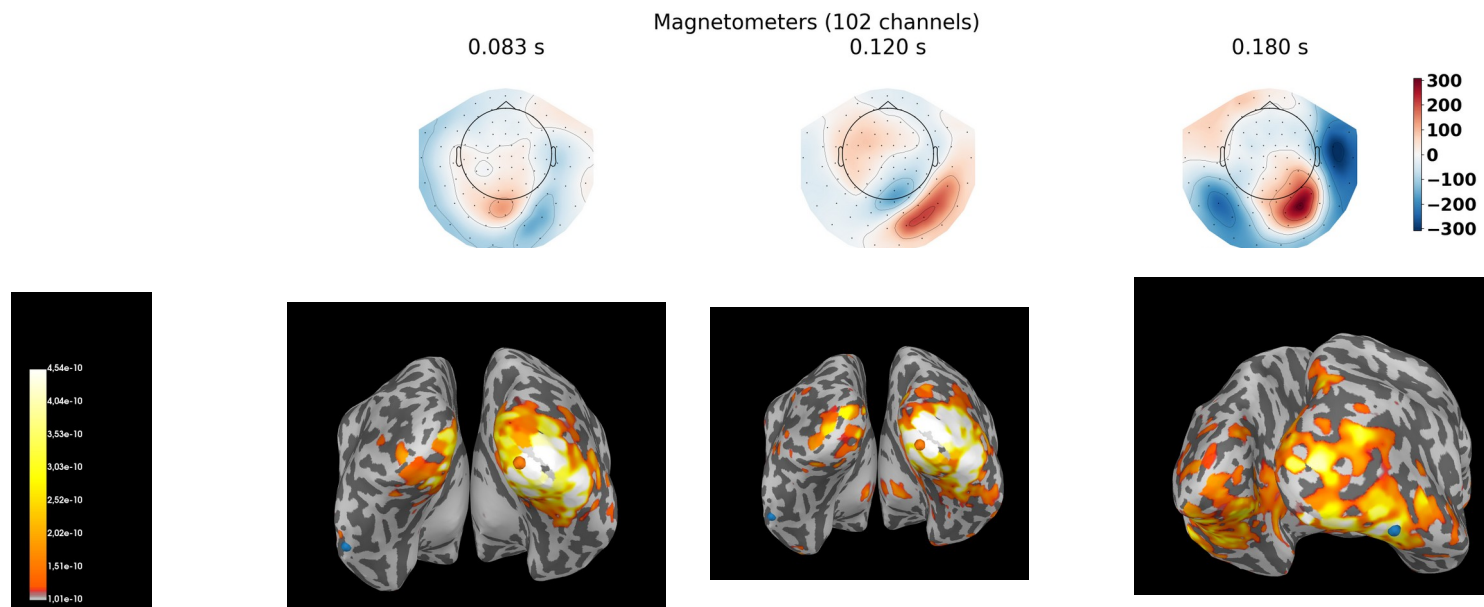
EVENT RELATED FIELD
(ERF)

Evoked responses to a left hemifield checkerboard (sample data)

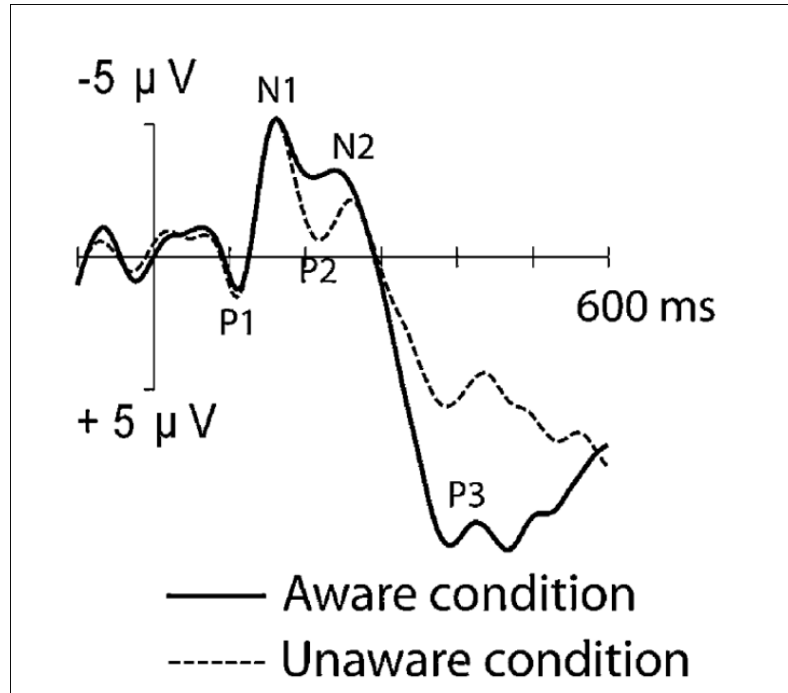
By Reddi - Own work, CC BY-SA 3.0, <https://commons.wikimedia.org/w/index.php?curid=2207791>



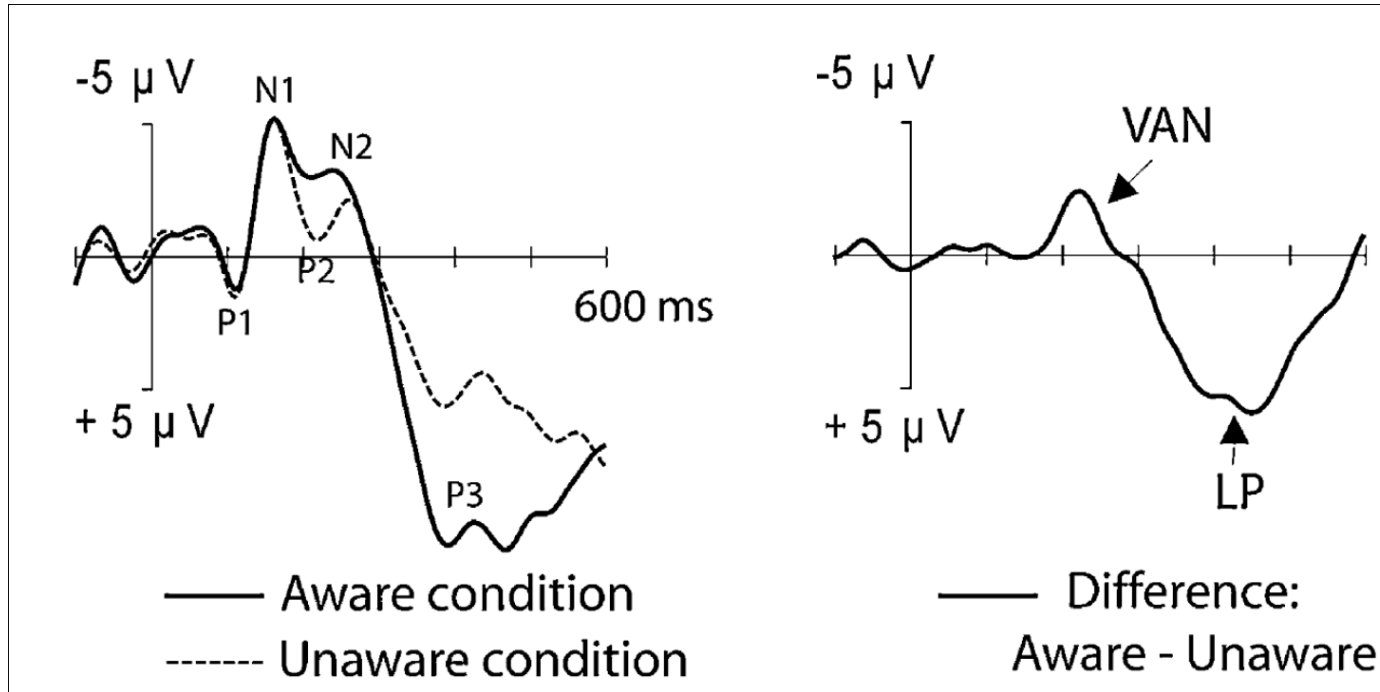
Evoked responses to a left hemifield checkerboard (sample data)



Early and late components (EEG)

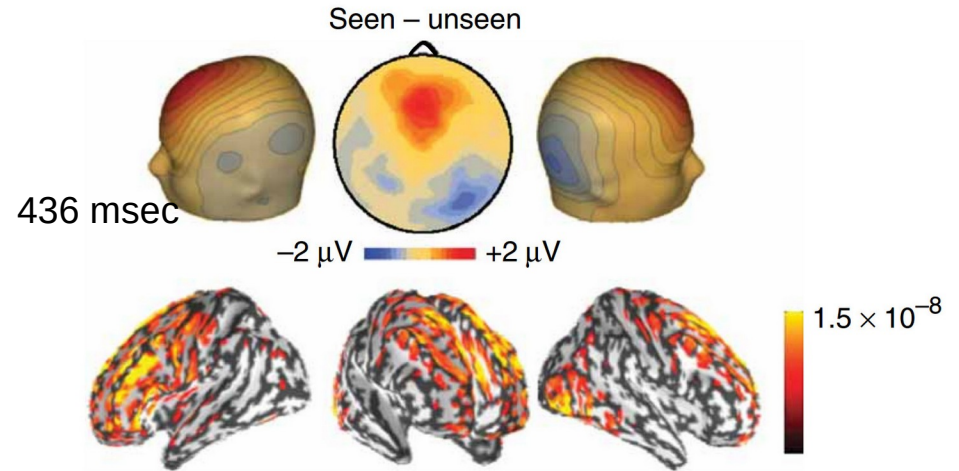
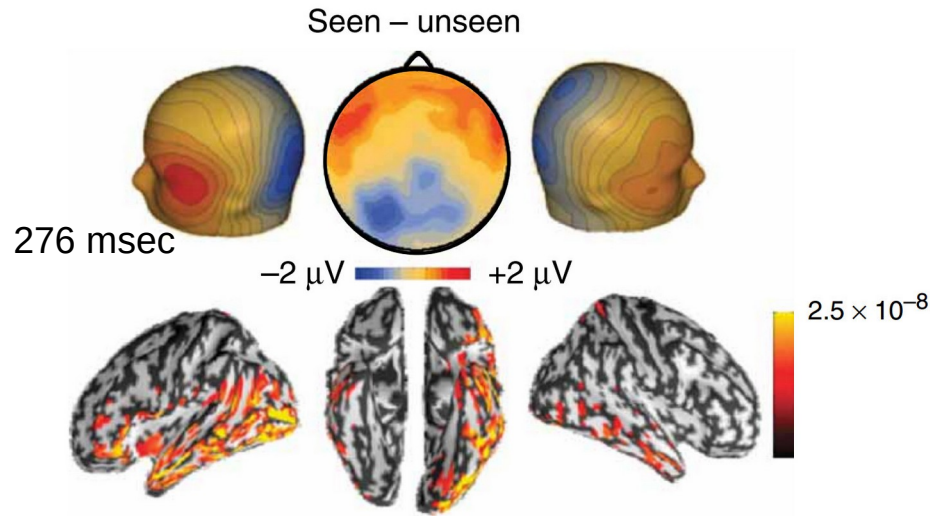


Early and late components (EEG)



Koivisto & Revonsuo 2010, *Neuroscience & Biobehavioral Reviews*

Early and late components and their source reconstructions (EEG)



Sergent et al. 2005, *Nature Neuroscience*

Where and when?

Consciousness of
visual content
happens *early* (~270
ms) and is *occipito-
temporally* realized

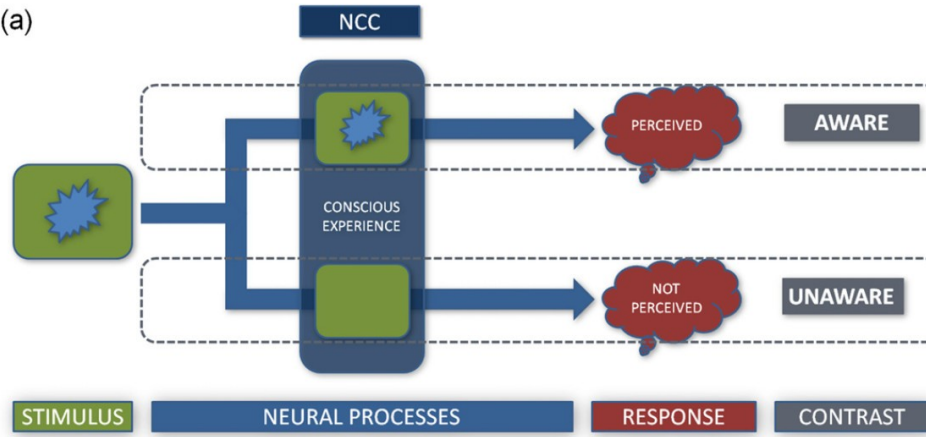
Where and when?

Consciousness of visual content happens *early* (~270 ms) and is *occipito-temporally* realized

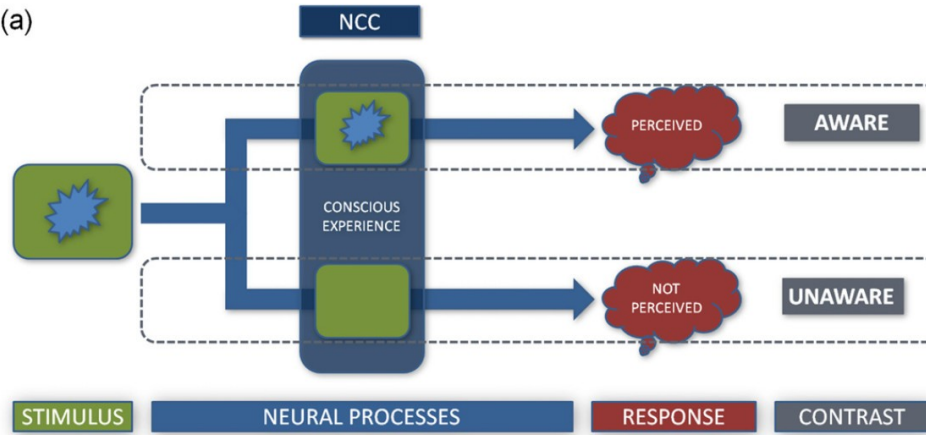
Consciousness of visual content happens *late* (~430 ms) and is *fronto-parietally* realized.

Which is the real neural
correlate of consciousness
(NCC)?

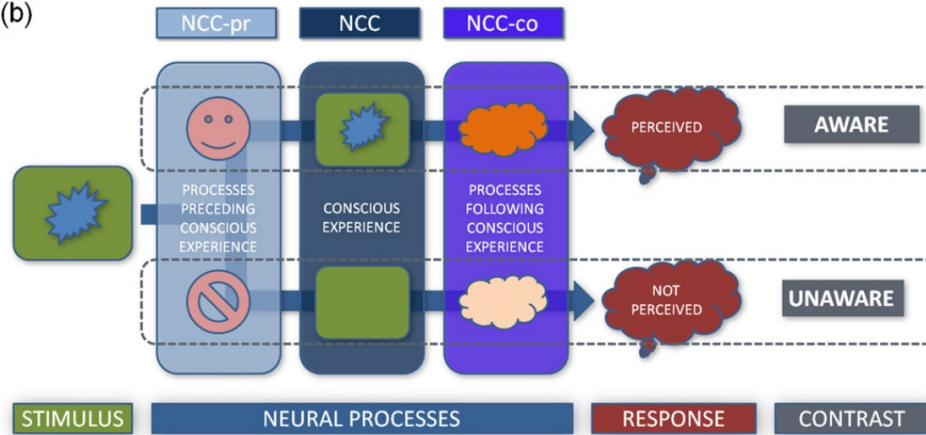
(a)



(a)



(b)



Aru et al. 2012, *Neuroscience & Biobehavioral Reviews*

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Where and when?

Consciousness of visual content happens *early* (~270 ms) and is *occipito-temporally* realized.

Consciousness of visual content happens *late* (~430 ms) and is *fronto-parietally* realized.

Where and when?

Consciousness of visual content happens *early* (~270 ms) and is *occipito-temporally* realized. The late activity is simply a consequence of consciousness.

Consciousness of visual content happens *late* (~430 ms) and is *fronto-parietally* realized.

Where and when?

Consciousness of visual content happens *early* (~270 ms) and is *occipito-temporally* realized. The late activity is simply a *consequence* of consciousness.

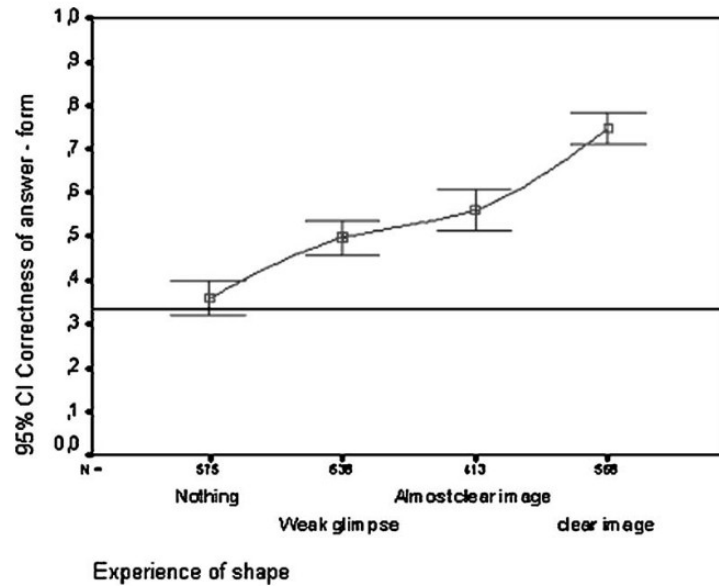
Consciousness of visual content happens *late* (~430 ms) and is *fronto-parietally* realized. The early activity is simply a *prerequisite* for consciousness.

Is subjective experience dichotomous or graded?

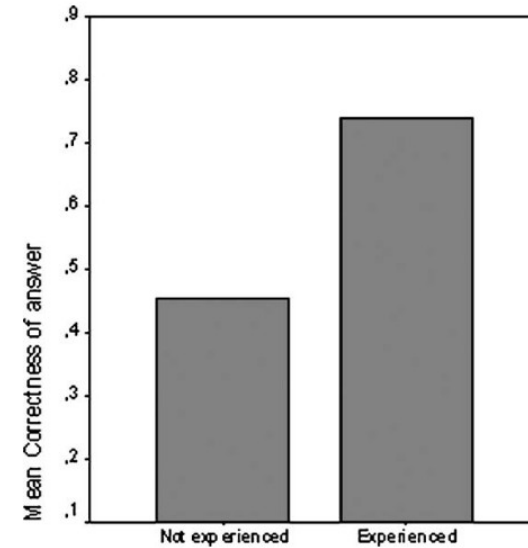
(Ramsøy and Overgaard 2004)



INTROSPECTION AND SUBLIMINAL PERCEPTION



INTROSPECTION AND SUBLIMINAL PERCEPTION



(Ramsøy and Overgaard 2004)

Exhaustiveness and sensitivity

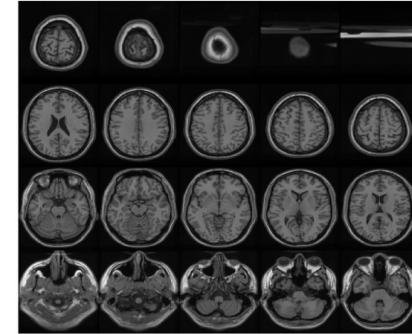
- Exhaustiveness
 - No more scale points are needed to capture the range of subjective experiences
- Sensitivity
 - The scale points given all reflect a real difference in behaviour

Interim summary

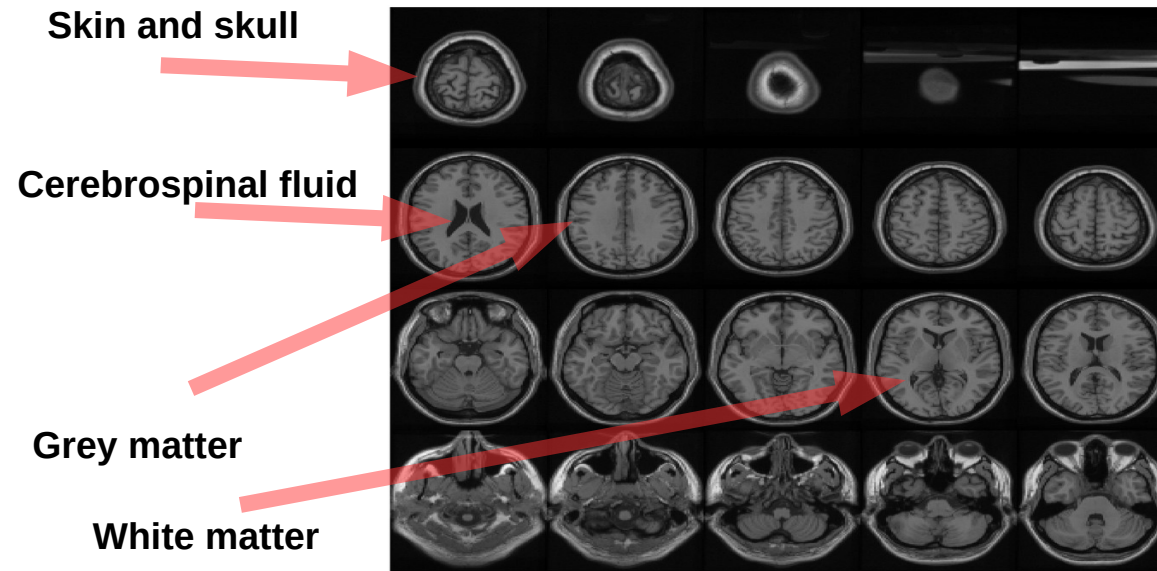
- Goal: find evidence for *where* and *when* subjective experience can be predicted the best
- Goal: find evidence, behavioural and electrophysiological, for the sensitivity and the exhaustiveness of the Perceptual Awareness Scale

What about MR?

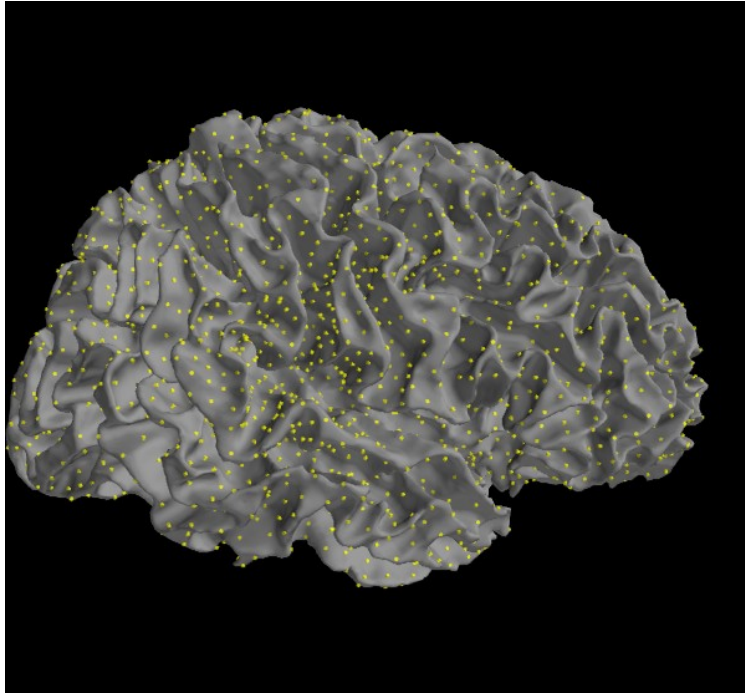
We need images of the brain to get estimates of brain activity



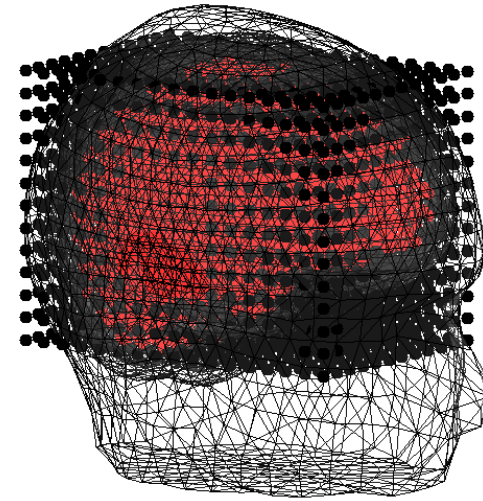
What do we see?



Source model examples



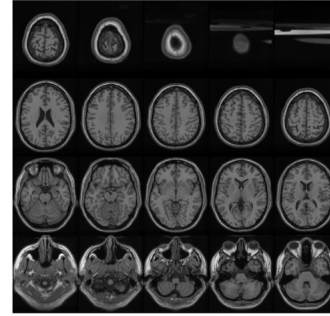
Restricted to the cortical surface



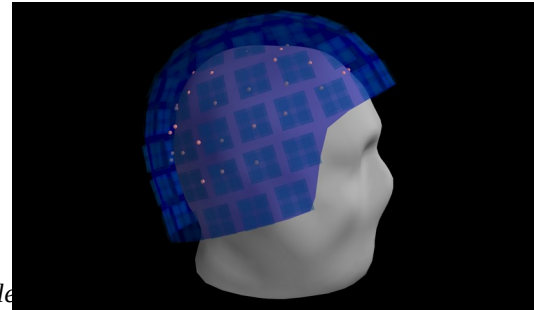
Volumetric grid

Sensor positions

- The MR-images and the helmet/electrode data will be represented in different coordinate systems
 - Thus, they have to be co-registered



≠



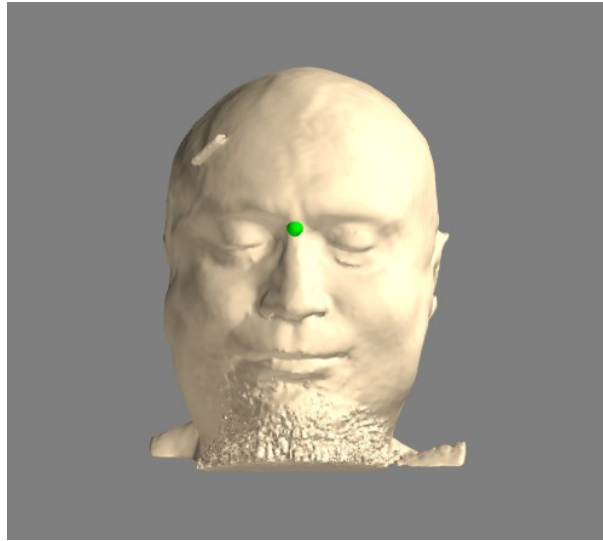
Co-registration



Construct a head model based on the MRI

Co-registration

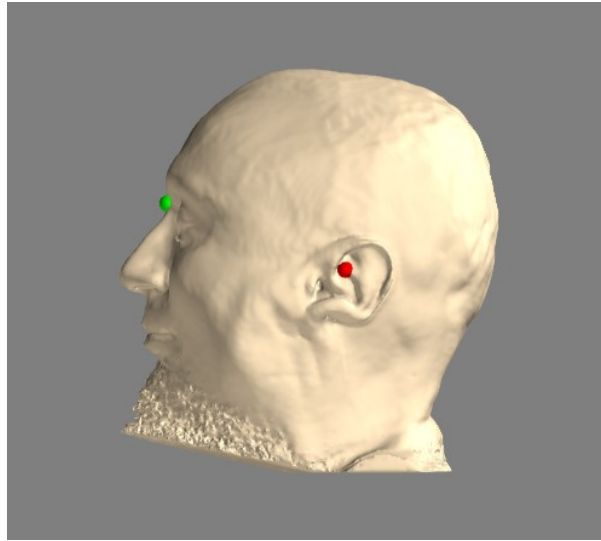
1. Nasion



Plot in fiducial points (these are acquired before recording)

Co-registration

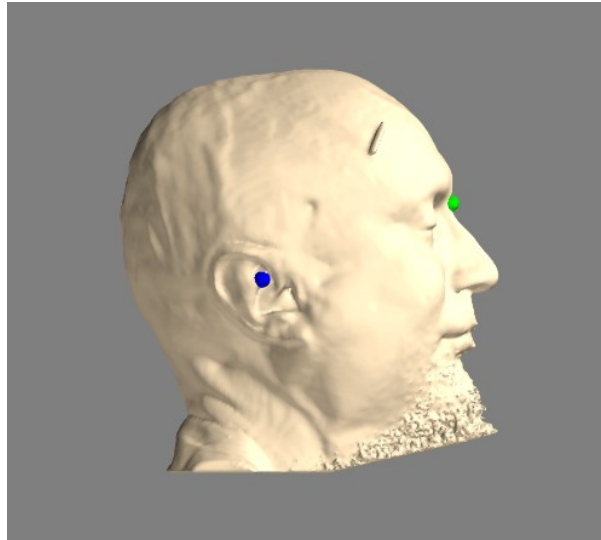
1. Nasion
2. Left pre-auricular point



Plot in fiducial points (these are acquired before recording)

Co-registration

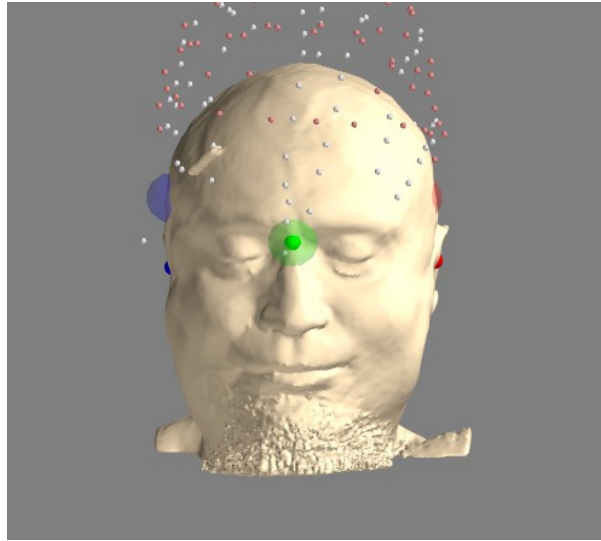
1. Nasion
2. Left pre-auricular point
3. Right pre-auricular point



Plot in fiducial points (these are acquired before recording)

Co-registration

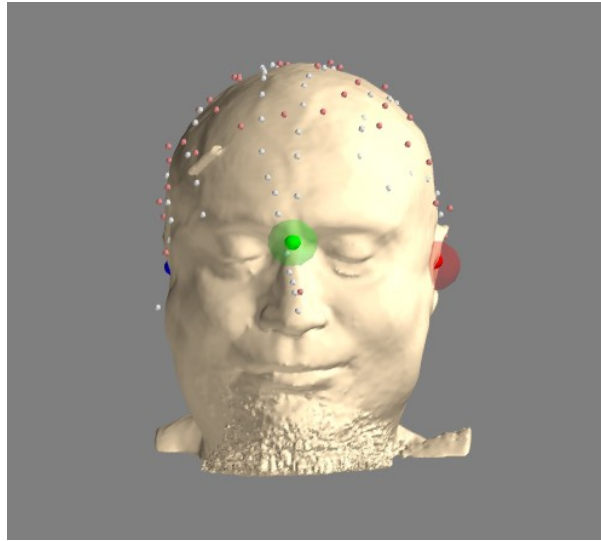
1. Nasion
2. Left pre-auricular point
3. Right pre-auricular point



Plot extra head points (as seen on lab tour)

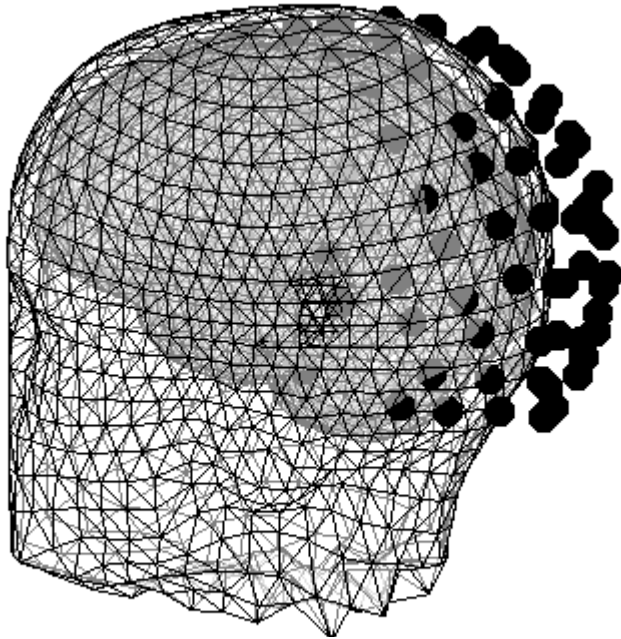
Co-registration

1. Nasion
2. Left pre-auricular point
3. Right pre-auricular point

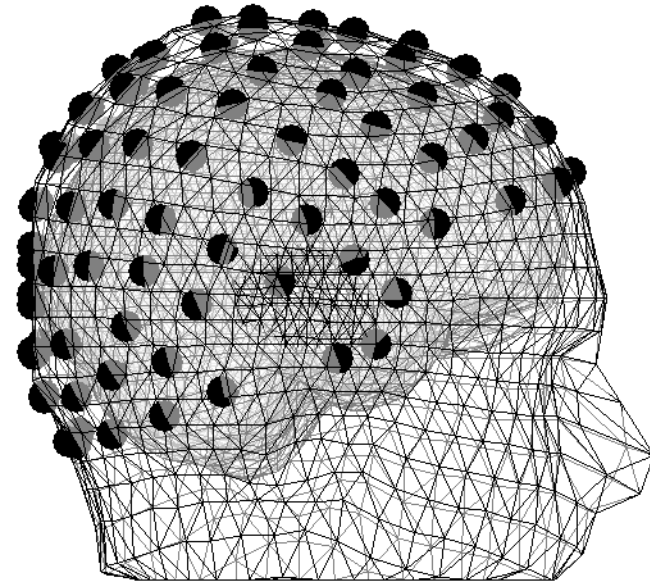


Use an algorithm to minimize the distance between points and head shape. The MR and the MEG are now co-registered

Without co-registration



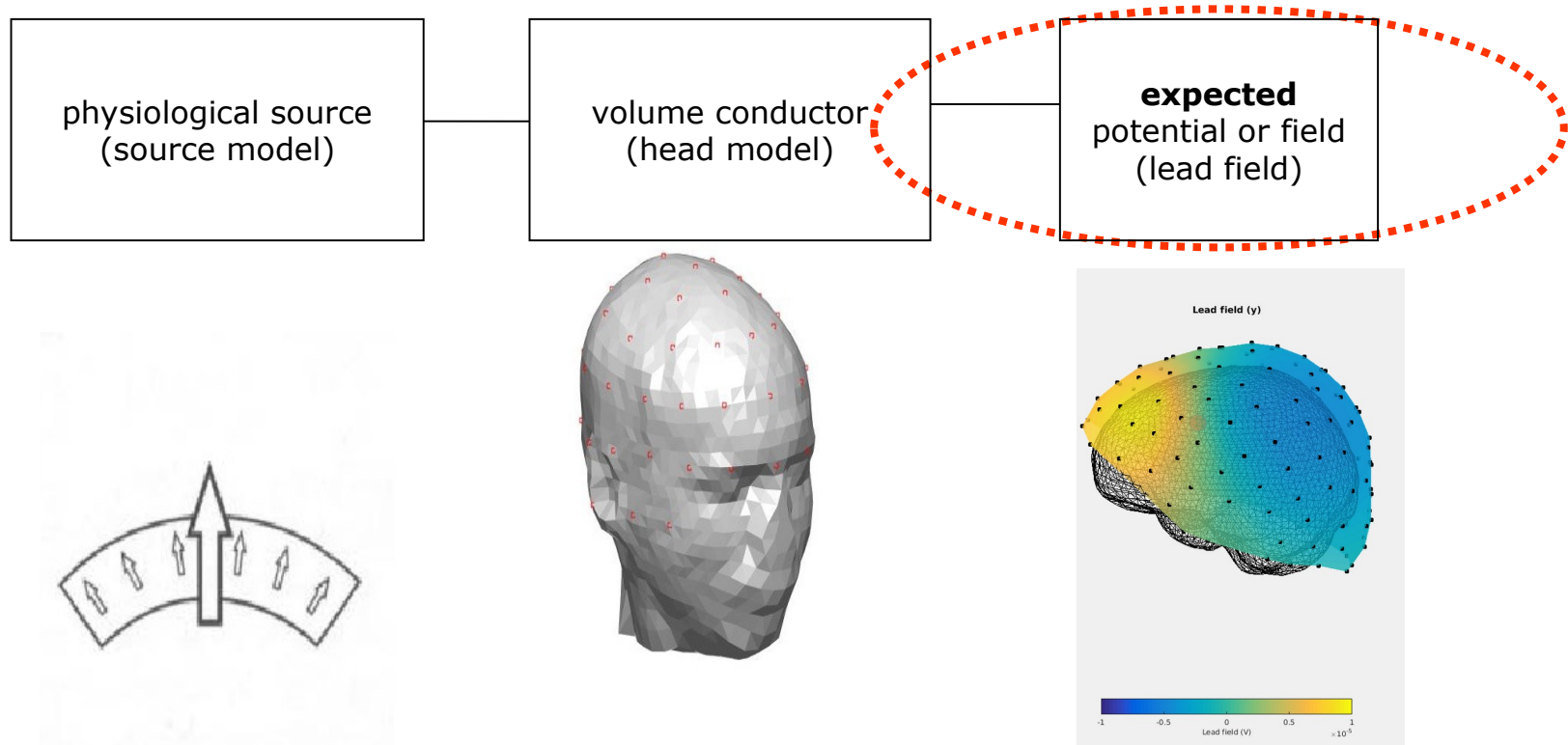
With co-registration

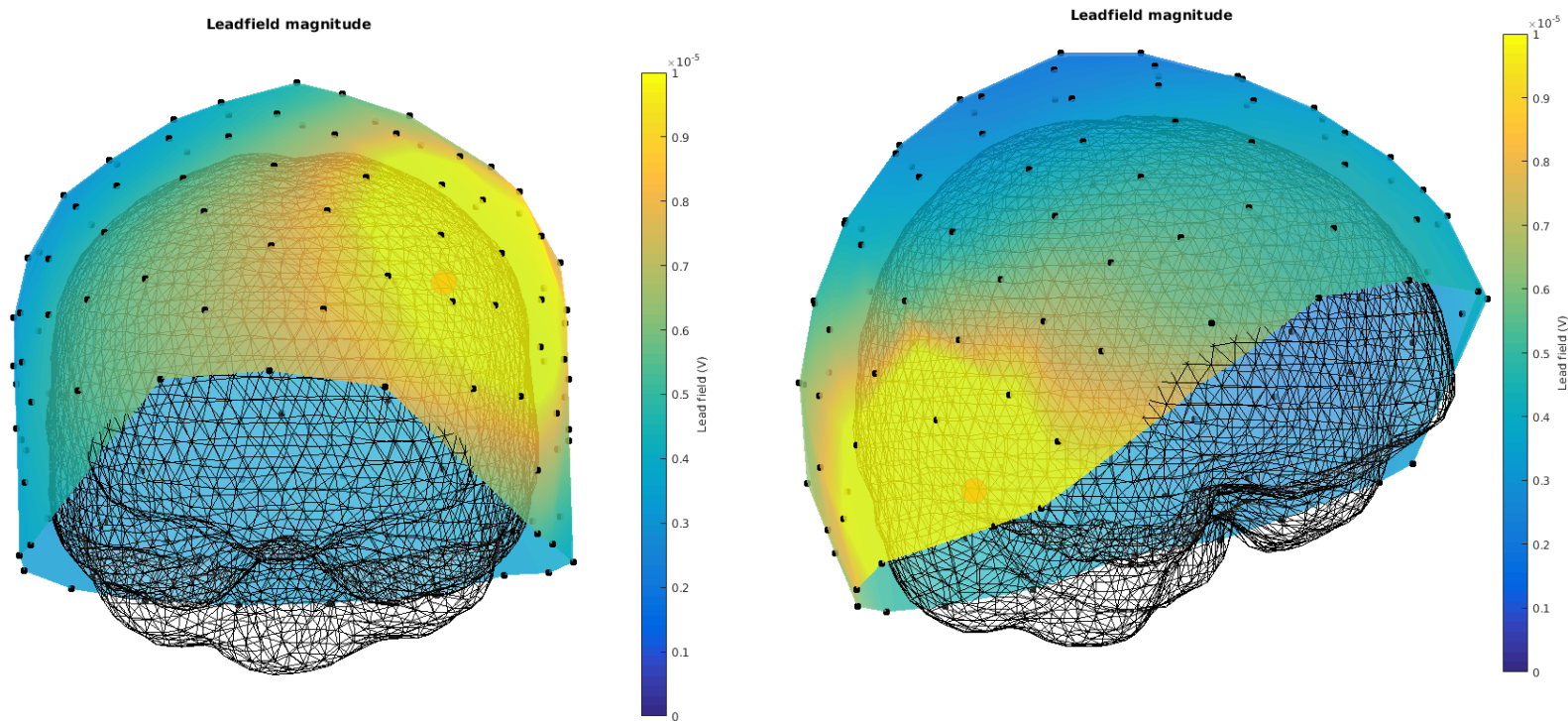


Now we can create the forward model (leadfield)

The forward model models how each source in the source space *would* be seen by the sensors in the helmet, *given* that that source is active

forward model

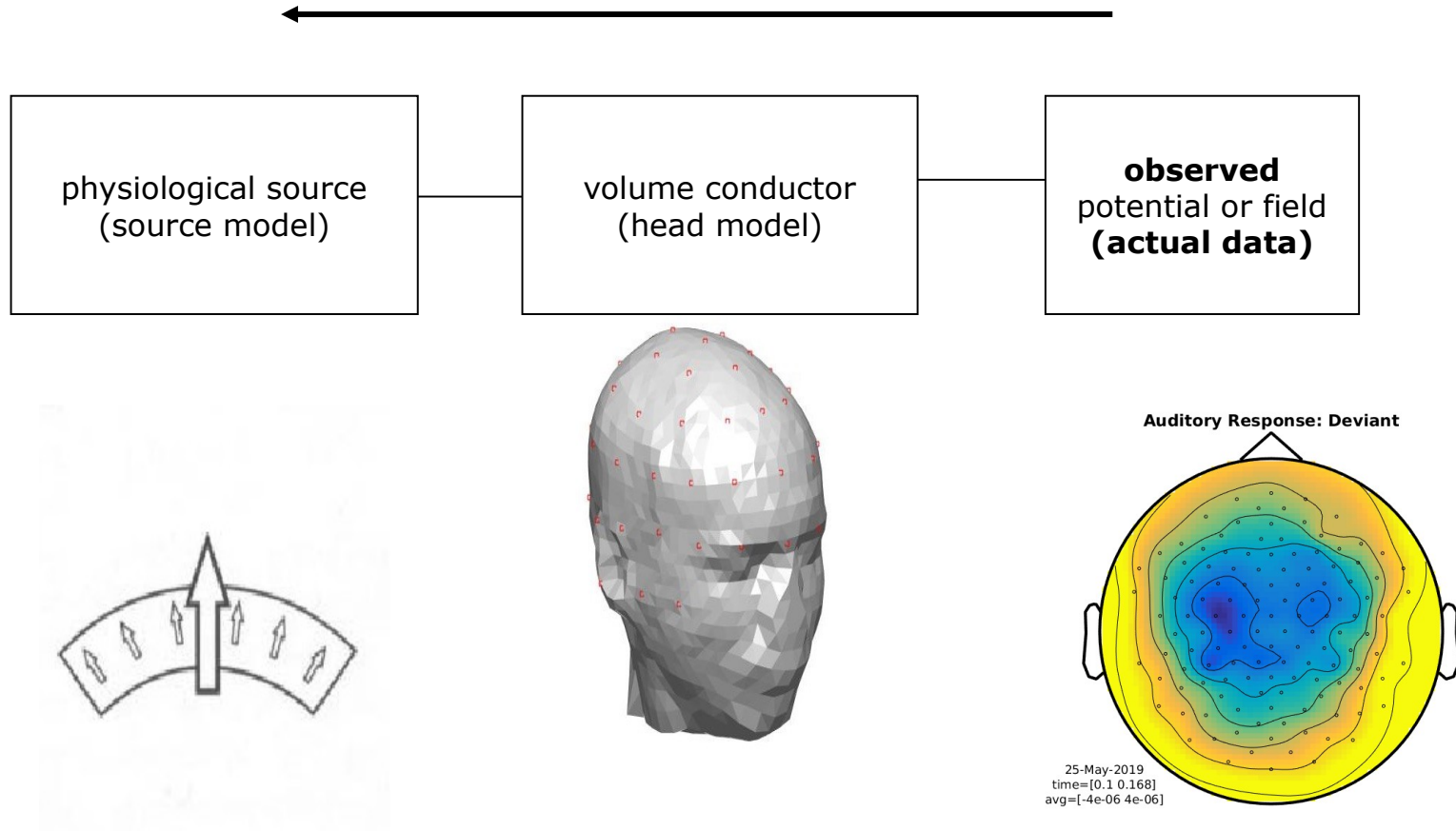




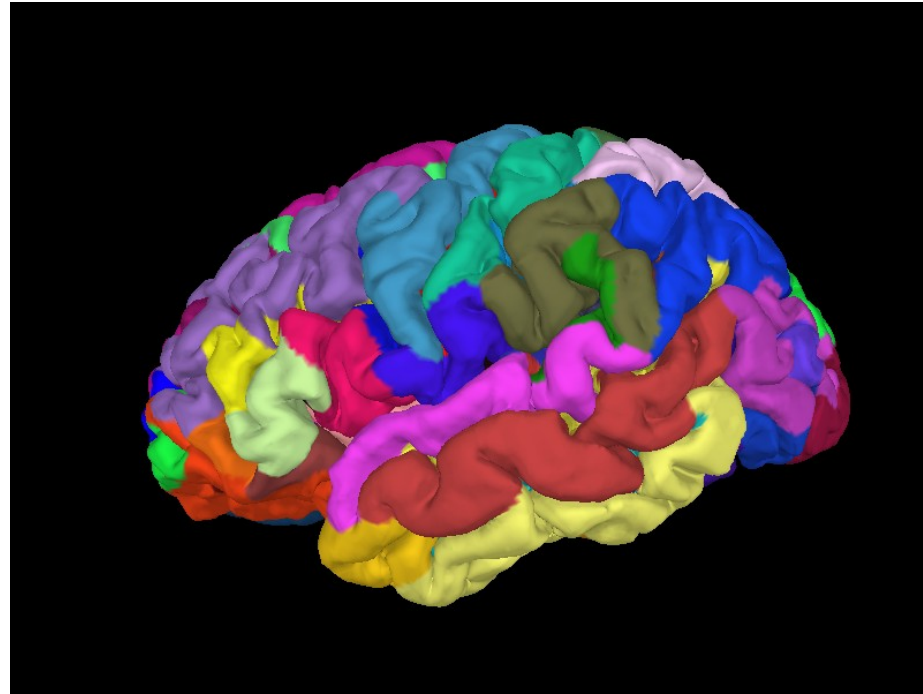
and so on for each source in the source model ...

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inverse model



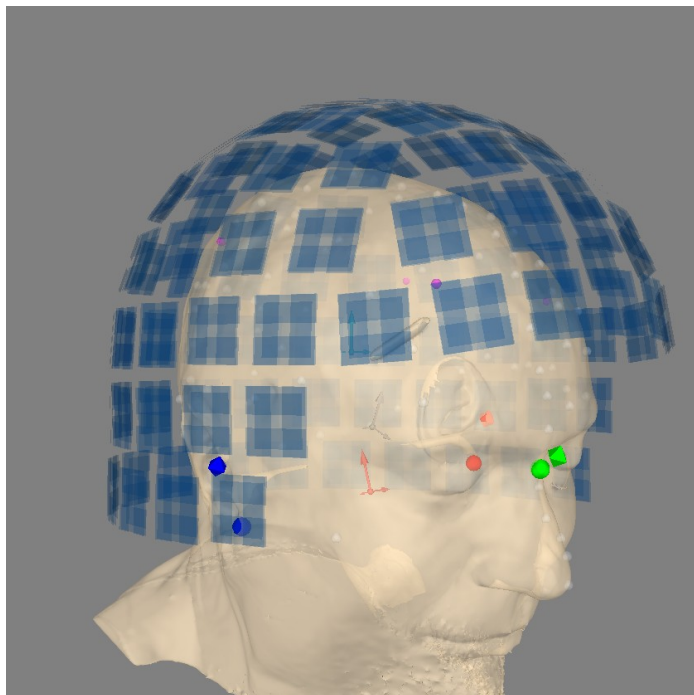
Segmented cerebral cortex



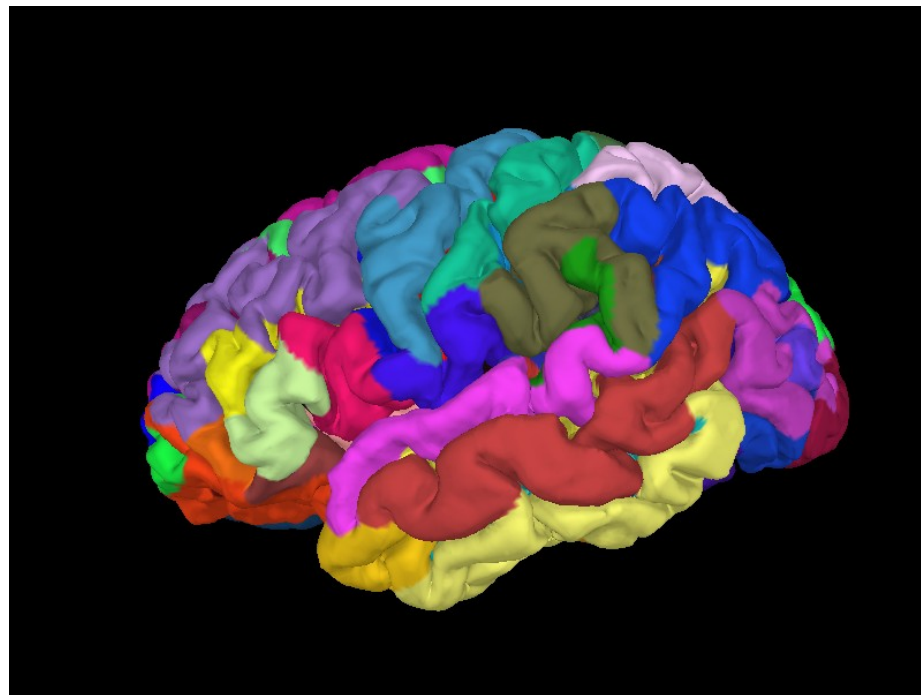
https://mne.tools/stable/auto_tutorials/forward/10_background_freesurfer.html#sphx-glr-auto-tutorials-forward-10-background-freesurfer-py

Multivariate analyses

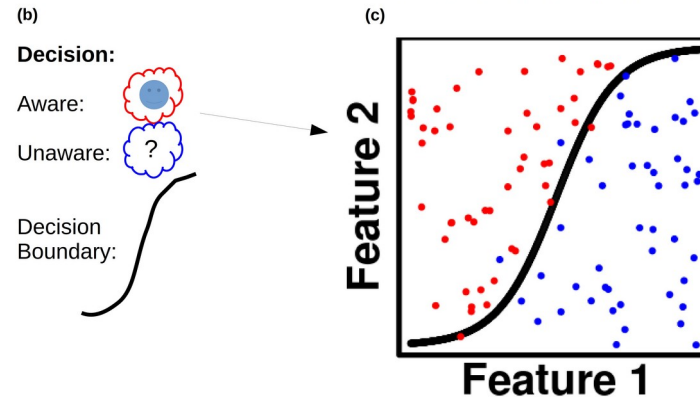
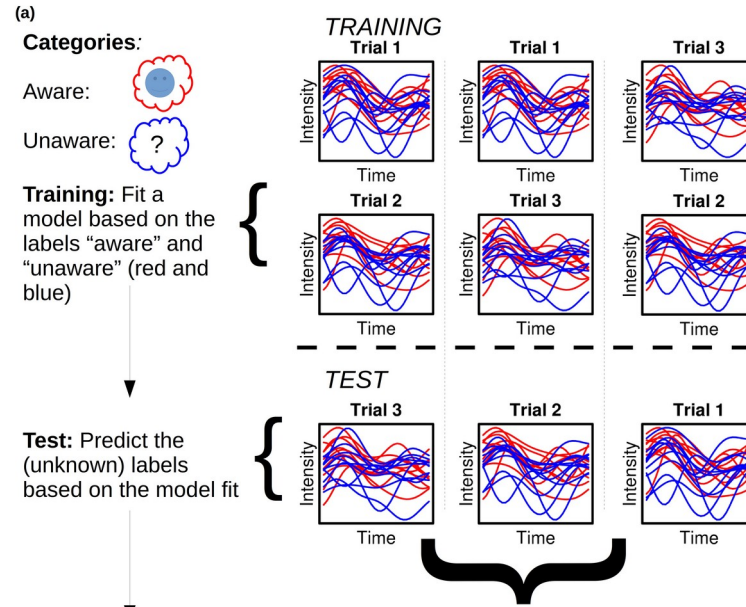
https://mne.tools/stable/auto_tutorials/forward/20_source_alignment.html#sphx-glr-auto-tutorials-forward-20-source-alignment-py



Sensor space



Source space



Pros and cons

- Sensor space
 - Likely to get better predictions
 - Lower specificity – where is the activity coming from?
- Source space
 - Likely to get worse predictions
 - Better specificity – we model where the activity is coming from

Summary

- Goal: find evidence for *where* and *when* subjective experience can be predicted the best
 - which can be informed by source space and brain segmentation
- Goal: find evidence, behavioural and electrophysiological, for the sensitivity and the exhaustiveness of the Perceptual Awareness Scale
 - You can use *glmer* to fit the proportion correct

The next three days: – workshop

Reading questions

- Chapters 5-7, Puce A, Hari R (2023)
 - Chapter 5
 - What are the differences between magnetometers and (planar) gradiometers?
 - Chapter 6
 - What is visual angle (see chapter 14?)
 - Chapter 7
 - What are fiducial points – and why are they important?
 - Why would you record other physiological signals, e.g. the heart and the eyes

Next class – Running an MEG analysis of visual responses